

# MetAgent case study

Input metabolite set -> tool-grounded pathway reasoning -> final metabolic interpretation

compound\_only\_enrich\_mammalian\_lm\_pathway\_WP167\_seed5

## Input

Differential metabolites (n=7)

### 5-HETE

C20H32O3

C04805

### 15-HETE

C20H32O3

C04742

### Leukotriene A4

C20H30O3

C00909

### 15-Deoxy-d-12,14-PGJ2

C20H28O3

C14717

### Arachidonic acid

C20H32O2

C00219

### Other differential metabolites

Glycolaldehyde

C00266

Losartan

C07072

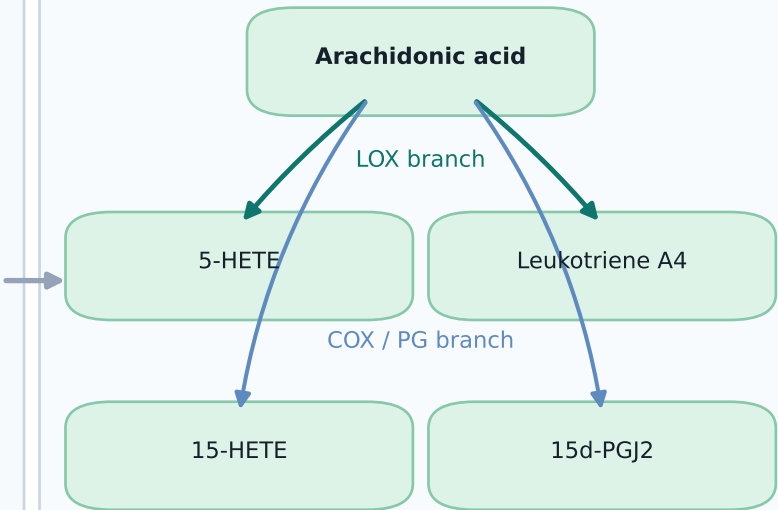
- Matched to top pathway
- Included in input, not pathway drivers

## Tool-grounded pathway evidence

### Top enrichment

### Eicosanoid synthesis

lipidmaps WP167 | FDR=2.47e-08 | matched 5/7



MetAgent used enrichment, pathway membership, KEGG path queries, and literature search to link the observed lipids to an arachidonic-acid eicosanoid cascade.

## Final MetAgent metabolic report

Representative claims from the selected final narrative

### SUPPORTED

biological\_claim

Eicosanoid synthesis is the top candidate pathway

### UNSUPPORTED

biological\_claim

Elevated HETEs in the treatment group are consistent with lipoxygenase-driven oxidation of arachidonic acid

### CONTRADICTED

consistency\_claim

Intra-document contradiction across claims [34], [35]

### UNVERIFIABLE\_V0

factual\_roundtrip\_claim

Arachidonic acid has KEGG ID C00219

### Final verifier totals

supported 13 | unsupported 3 | contradicted 1  
unverifiable\_v0 35